

DOUGLAS MENDES BARRETO

Staff Software Engineer

+1 (581) 748 0657 • mendes-barreto@live.com • [LinkedIn](#) • [GitHub](#) • [Portfolio](#) • Gatineau, QC - Canada

Professional Summary

Staff-level software engineer with **14+ years** of experience building products across backend, frontend, mobile, cloud infrastructure, and applied AI. I operate at the intersection of architecture, execution, and cross-functional leadership, helping teams turn ambiguous product goals into reliable systems and shipped outcomes.

My work is centered on scaling engineering through better technical direction, stronger platform foundations, practical automation, and clear decision-making. I stay hands-on with architecture, critical implementation work, and delivery strategy, especially in environments that require modernization, migration, or multi-team coordination.

I'm most effective in roles that need broad technical range, end-to-end ownership, and engineering leadership without losing proximity to the code.

Staff Engineering Strengths

- Driving technical strategy across product, platform, and infrastructure
- Leading architecture decisions and translating them into executable roadmaps
- Operating as a hands-on technical lead on high-leverage and high-ambiguity initiatives
- Improving delivery systems, developer experience, and engineering effectiveness
- Leading migrations, modernization efforts, and platform evolution across multiple domains
- Partnering with product, design, data, and operations to align technical choices with business outcomes
- Mentoring senior engineers and raising engineering standards through design review and technical guidance
- Building scalable systems using pragmatic tradeoffs instead of overengineering
- Introducing automation and AI-assisted workflows that accelerate prototyping and delivery
- Communicating clearly with stakeholders through technical narratives, proposals, and execution plans

Tech Skills

- Languages: Python, JavaScript/TypeScript, Go, C++, C#, Ruby, Swift, Objective-C
 - Backend and APIs: Node.js, GraphQL, REST APIs, microservices, event-driven systems
 - Frontend and Mobile: React, Next.js, React Native, native iOS, native Android
 - Cloud and Platform: Google Cloud, AWS, Docker, Kubernetes, Terraform, Ansible, CI/CD
 - Data and Storage: MySQL, PostgreSQL, MongoDB, Redis
 - Engineering Practices: system design, testing strategy, trunk-based development, TDD/BDD, design patterns, SOLID
 - Tooling and Automation: GitHub Actions, Jenkins, build pipeline optimization, developer workflows, AI-powered development tools
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Professional Experience

Senior Software Engineering Manager / Staff-Level Technical Lead

Xpertsea - Predictive shrimp biomass control with AI delivering sustainable aquaculture solutions

51 - 100 employees

Quebec, Canada

Oct 2020 – Present

- Led technical direction for a product portfolio spanning AI-assisted workflows, computer vision, predictive systems, and operational software
 - Served as the senior engineering lead for a 9-engineer team, combining people leadership with hands-on architecture and delivery ownership
 - Built a self-service AI prototyping workflow that materially reduced time required for early product validation
 - Increased delivery speed by 5x, reducing lead time from 13 weeks to 2 weeks through workflow, architecture, and process changes
 - Introduced Kanban and trunk-based development to improve flow, reduce batch size, and strengthen release cadence
 - Reworked CI/CD pipelines, reducing build and deploy time from 50 minutes to 7 minutes
 - Led a full-stack migration from AWS to Google Cloud, reducing infrastructure costs by 20% while improving platform consistency
 - Drove platform improvements that strengthened scalability, developer experience, and cross-team execution
 - Partnered with product and design to align roadmap decisions with technical constraints, business goals, and delivery capacity
 - Supported engineers through architecture guidance, technical review, and high-leverage implementation across backend, frontend, and computer vision work
 - Oversaw the successful launch of 6 major projects
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Senior iOS Developer

Social.mom - Social Media for mothers focused on delivering a safe and supportive environment

11 - 20 employees

Oct 2019 – Oct 2020

- Led re-architecture of a high-traffic iOS application, reducing maintenance overhead and cutting feature development time by 20%
 - Drove migration of key features from Objective-C to Swift, improving codebase consistency and maintainability
 - Built automation scripts that removed hundreds of hours of repetitive manual work from the delivery process
 - Introduced unit and end-to-end testing, increasing release confidence and improving product reliability
 - Acted as a senior technical reference on architecture, code quality, and modern engineering practices
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Senior Software Engineer

Levio - Tech Consulting

1K - 5K employees

Jul 2018 – Oct 2019

- Built and maintained high-availability backend services for multiple clients using .NET C#
 - Led legacy modernization efforts, migrating critical systems from COBOL to modern .NET platforms
 - Improved collaboration and delivery practices by guiding migrations from legacy version control systems to Git
 - Worked directly with client stakeholders and cross-functional teams to shape requirements and deliver robust systems
 - Optimized CI pipelines to support faster and more reliable delivery
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Senior Mobile Engineer

Concrete Solutions - Tech Consulting

10K+ employees

Jul 2015 – Aug 2018

- Led architecture and development of native mobile applications across financial and retail clients
 - Delivered 3 mobile banking applications and 4 e-commerce apps in regulated and high-scale environments
 - Architected a white-label e-commerce mobile platform that enabled rapid rollout of customized storefronts from one codebase
 - Designed CI/CD pipelines with Jenkins, Bitrise, and Travis CI to improve release reliability and team throughput
 - Raised engineering quality through technical talks on coding standards, testing practices, and QA implementation
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Lead Game Programmer

Editora Globo - Publisher and developer of digital entertainment products

500 - 1K employees

Aug 2010 – Apr 2015

- Led end-to-end engineering of cross-platform games for iOS, Android, and Windows using C# and Unity3D
 - Architected backend services for MMO-style games, including authentication, room control, and persistent multiplayer systems
 - Built reusable Unity3D framework components and tooling that improved team productivity and consistency
 - Developed gameplay and platform systems for products combining educational goals with large-scale digital distribution
 - Helped formalize planning and execution processes through JIRA adoption
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Junior Analyst Programmer

People Way - Tech Consulting

11 - 50 employees

Apr 2009 – Feb 2010

- Developed enterprise web applications using J2EE, JSP, JSF, and SOAP web services
 - Designed MySQL-backed data workflows and reporting integrations with JasperReports
 - Built parsers and automation scripts for SMS, XML, and CSV data processing
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Education

University Anhembi Morumbi

Bachelor's Degree in Game Design and Development - 2010

Languages

- English: Fluent
- Portuguese: Native
- French: Professional

Main Projects

Kampi (Xpertsea)

Intelligent copilot for shrimp farmers using AI and computer vision to replace manual guesswork with data-driven insights. Features three pillars: Pond Vision for biometric monitoring via smartphone photos, Optima for predictive analytics and growth forecasting, and FarmBook for team coordination across the production chain.

Tech Stack: Python, TypeScript/JS, MongoDB, MySQL, Node.js, Bun, Next.js, Google Cloud, Terraform, Ansible, C++, Android, iOS, React Native + Native Development

Link: <https://bekampi.com/>

- Led technical direction for an AI-powered product spanning computer vision, predictive analytics, and operational workflows
 - Defined architecture and execution strategy for predictive systems and computer vision capabilities
 - Guided delivery of an omnichannel platform supporting iOS, iPadOS, and web experiences
 - Directed infrastructure modernization using Google Cloud, Terraform, and Ansible
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Social.Mom (Social.mom)

Social network platform designed to counter social isolation during motherhood and reinforce the social safety net of families. Enables mothers to connect locally, plan activities, and access community groups and organizations in their area through a bilingual interface.

Tech Stack: iOS native, Android native, Java Spring Boot, React

Link: <https://social.mom/en>

- Led major iOS architecture work for a high-traffic bilingual social platform
 - Reworked native foundations to reduce maintenance overhead and improve delivery speed
 - Established testing practices that improved reliability and release confidence
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Meu Carrefour (Carrefour Brazil)

All-in-one retail companion and cornerstone of Carrefour Brazil's omnichannel strategy, empowering millions of users to manage loyalty rewards, shop for groceries online, and streamline in-store visits through smart utility tools like barcode scanning and proximity services.

Tech Stack: iOS native, TypeScript, Node.js, Java

- Built core mobile commerce experiences, including barcode scanning and store locator capabilities
 - Integrated loyalty systems with discount and cashback tracking
 - Implemented synchronized shopping list workflows across platforms
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Itaú Unibanco Mobile Banking (Itaú Unibanco)

Mobile banking application for one of Brazil's largest financial institutions, providing users with secure access to core banking services including account management and financial transactions.

Tech Stack: Windows Phone, iOS, .NET

- Delivered secure authentication and account access workflows for a large-scale banking product
 - Built transaction visibility features including account history and balances
 - Implemented transfers and payments as part of the core banking experience
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Mundo do Sítio (Editora Globo)

Persistent, multi-user virtual world designed to integrate pedagogical goals with immersive gameplay as a digital extension of the Sítio do Picapau Amarelo IP. Featured complex economy, social interactions, and library of educational mini-games for millions of Brazilian students.

Tech Stack: Unity 3D, Flash ActionScript (SWC), Java, SmartFox Server, HTML5/JS

- Architected full-stack systems for high-concurrency traffic and low-latency multiplayer communication
 - Developed gameplay systems and real-time state synchronization for 30+ educational mini-games
 - Implemented authentication, profile registration, and parental controls for a children's platform
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